

A Tangent-Binomial Representation for Complex Exponentials: Properties, Convergence, and Applications

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Abstract

The representation of complex exponentials is fundamental across mathematics, physics, and engineering. This paper presents a comprehensive analysis of the tangent-binomial representation of e^{inz} , which follows directly from De Moivre's theorem but whose properties are not widely explored. We formally derive this representation, $e^{inz} = \cos^n z (1 + i \tan z)^n$, and investigate its mathematical structure. Our primary contributions include a rigorous analysis of the associated tangent polynomials and their roots, a formal proof of the convergence criteria for the generalization to non-integer exponents $n \in \mathbb{C}$, and a discussion of its behavior at the boundary of convergence. We establish that the infinite series representation converges absolutely for $|\tan z| < 1$. Furthermore, we propose a framework for evaluating the numerical stability and computational efficiency of this representation compared to standard library implementations, particularly in domains where $\tan z$ is small. This work positions the tangent-binomial form as a valuable pedagogical tool and a potentially useful representation for specific computational applications.

1 Introduction

1.1 Background and Motivation

Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, is a cornerstone of modern mathematics, providing a profound link between exponential functions and trigonometry. Its extension to integer powers via De Moivre's theorem, $(\cos z + i \sin z)^n = \cos(nz) + i \sin(nz)$, is indispensable in complex analysis, wave mechanics, and signal processing [2]. While these formulas are canonical, the various algebraic forms they can take often reveal deeper structural properties or offer computational advantages in specific contexts. This paper focuses on one such form, which we term the tangent-binomial representation, arising from a simple factoring of the $\cos z$ term.

1.2 Literature Review and Contribution

The expression of $\cos(nz)$ and $\sin(nz)$ as polynomials in $\cos z$ and $\sin z$ is classical, leading to structures like the Chebyshev polynomials. The reformulation of these in terms of $\tan z$ is also a known procedure, often used in specific integration problems or theoretical derivations [1]. The core identity analyzed herein,

$$e^{inz} = \cos^n z (1 + i \tan z)^n,$$

is a direct and elementary consequence of De Moivre's theorem. However, a dedicated study of its properties—particularly its generalization to complex exponents and its performance as a numerical algorithm—is not prominent in the literature.

The contribution of this paper is therefore not the discovery of a new identity, but rather a rigorous and systematic investigation of this particular representation. We aim to:

1. Formally define and analyze the "tangent polynomials" that arise from this form.
2. Provide a rigorous proof of the convergence conditions for its generalization to $n \in \mathbb{C}$.
3. Establish a framework for testing its practical utility in numerical computation.

1.3 Paper Structure

The paper is organized as follows. Section 2 details the derivation of the tangent-binomial representation and its connection to hyperbolic functions. Section 3, the core of our theoretical contribution, provides a detailed analysis of the mathematical properties of the representation, including its polynomial structure and convergence for non-integer exponents. Section 4 discusses potential applications and outlines a protocol for numerical analysis. Finally, Section 5 concludes with a summary of our findings and suggests avenues for future research.

2 The Tangent-Binomial Representation

In this section, we formally derive the central identity of this paper. We begin with a motivating, non-standard derivation before presenting the more direct proof.

2.1 A Motivating Derivation using a Tangent-Secant Approach

We begin with Euler's formula for a complex argument z :

$$e^{iz} = \cos z + i \sin z \quad (1)$$

Assuming $\cos z \neq 0$, we can divide both sides by $\cos z$:

$$\frac{e^{iz}}{\cos z} = 1 + i \frac{\sin z}{\cos z} \quad (2)$$

Using the definitions $\sec z = 1/\cos z$ and $\tan z = \sin z/\cos z$, we arrive at the compact relation:

$$e^{iz} \sec z = 1 + i \tan z \quad (3)$$

This identity elegantly connects the exponential to the tangent and secant functions. Raising both sides to the power of an integer n immediately suggests a more general form. While this approach is insightful, a more direct derivation is preferable for a formal proof.

2.2 General Formulation via De Moivre's Theorem

We now present the direct and rigorous derivation.

Theorem 2.1 (Tangent-Binomial Representation). *For any $n \in \mathbb{Z}$ and $z \in \mathbb{C}$ such that $\cos z \neq 0$, the complex exponential e^{inz} can be expressed as:*

$$e^{inz} = \cos^n z \sum_{k=0}^n \binom{n}{k} (i \tan z)^k \quad (4)$$

Proof. We begin with De Moivre's formula:

$$e^{inz} = (e^{iz})^n = (\cos z + i \sin z)^n$$

Factoring out the term $\cos z$, we obtain:

$$e^{inz} = \left[\cos z \left(1 + i \frac{\sin z}{\cos z} \right) \right]^n = \cos^n z (1 + i \tan z)^n$$

Applying the binomial theorem to the term $(1 + i \tan z)^n$, we get:

$$(1 + i \tan z)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} (i \tan z)^k = \sum_{k=0}^n \binom{n}{k} (i \tan z)^k$$

Substituting this back yields the desired result:

$$e^{inz} = \cos^n z \sum_{k=0}^n \binom{n}{k} (i \tan z)^k$$

This completes the proof. □

2.3 Connection to Hyperbolic Functions

The representation provides a clear pathway to standard hyperbolic identities. Let $z = iy$ for some $y \in \mathbb{R}$. Then e^{inz} becomes $e^{in(iy)} = e^{-ny}$. The trigonometric functions become:

$$\begin{aligned}\cos(iy) &= \cosh y \\ \sin(iy) &= i \sinh y \\ \tan(iy) &= i \tanh y\end{aligned}$$

Substituting these into the identity from Eq. (4) for $n = 1$:

$$\begin{aligned}e^{-y} &= \cos(iy)(1 + i \tan(iy)) \\ &= \cosh y(1 + i(i \tanh y)) \\ &= \cosh y(1 - \tanh y) \\ &= \cosh y \left(1 - \frac{\sinh y}{\cosh y}\right) \\ &= \cosh y - \sinh y\end{aligned}$$

This correctly recovers the well-known identity for the exponential function in terms of hyperbolic cosine and sine.

3 Mathematical Properties and Analysis

We now proceed to a deeper analysis of the tangent-binomial representation, exploring the structure of its constituent polynomials and its generalization to non-integer exponents.

3.1 The Associated Tangent Polynomials

We can define a family of complex polynomials in the variable u that are central to the representation.

Definition 3.1 (Associated Tangent Polynomials). For $n \in \mathbb{N}$, the n -th Associated Tangent Polynomial, $P_n(u)$, is defined as:

$$P_n(u) = \sum_{k=0}^n \binom{n}{k} i^k u^k = (1 + iu)^n$$

With this definition, Eq. (4) can be written as $e^{inz} = \cos^n z \cdot P_n(\tan z)$.

Proposition 3.1 (Roots of $P_n(u)$). The polynomial $P_n(u)$ has a single root of multiplicity n at $u = i$.

Proof. The roots of $P_n(u) = (1 + iu)^n = 0$ are the solutions to $1 + iu = 0$. This yields $iu = -1$, or $u = -1/i = i$. Since the factor $(1 + iu)$ is repeated n times, the root at $u = i$ has multiplicity n . $\square$

Remark. This result has a clear interpretation. The representation fails only when $P_n(\tan z)$ is undefined, which occurs when $\cos z = 0$. The roots of the polynomial itself do not introduce additional singularities into the representation for e^{inz} .

3.2 Generalization to Non-Integer Exponents

The representation can be formally extended to any exponent $n \in \mathbb{C}$ by replacing the finite sum with the generalized binomial series.

Definition 3.2 (Generalized Tangent-Binomial Representation). For any $n \in \mathbb{C}$ and $z \in \mathbb{C}$, we define the generalized representation as:

$$e^{inz} = \cos^n z \sum_{k=0}^{\infty} \binom{n}{k} (i \tan z)^k \quad (5)$$

where $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!}$ is the generalized binomial coefficient.

The validity of this expression depends on the convergence of the infinite series.

Theorem 3.2 (Convergence Condition). *The infinite series in Eq. (5) converges absolutely if and only if $|\tan z| < 1$.*

Proof. We apply the ratio test to the series $S = \sum_{k=0}^{\infty} a_k$, where $a_k = \binom{n}{k} (i \tan z)^k$. The ratio of successive terms is:

$$\begin{aligned} \left| \frac{a_{k+1}}{a_k} \right| &= \left| \frac{\binom{n}{k+1} (i \tan z)^{k+1}}{\binom{n}{k} (i \tan z)^k} \right| \\ &= \left| \frac{n(n-1) \cdots (n-k)}{(k+1)!} \cdot \frac{k!}{n(n-1) \cdots (n-k+1)} \right| \cdot |i \tan z| \\ &= \left| \frac{n-k}{k+1} \right| \cdot |\tan z| \end{aligned}$$

For the series to converge, we require the limit of this ratio as $k \rightarrow \infty$ to be less than 1:

$$L = \lim_{k \rightarrow \infty} \left| \frac{n-k}{k+1} \right| |\tan z|$$

Evaluating the limit of the rational part:

$$\lim_{k \rightarrow \infty} \left| \frac{n-k}{k+1} \right| = \lim_{k \rightarrow \infty} \left| \frac{n/k - 1}{1 + 1/k} \right| = \left| \frac{-1}{1} \right| = 1$$

Thus, the limit L simplifies to:

$$L = |\tan z|$$

According to the ratio test, the series converges absolutely if $L < 1$, which implies $|\tan z| < 1$. It diverges if $L > 1$, which implies $|\tan z| > 1$. $\square$

3.3 Analysis at the Boundary of Convergence

The case where $|\tan z| = 1$ requires a more delicate analysis (such as Raabe's test or Gauss's test) and depends on the specific value of n . For a general $n \in \mathbb{C}$, behavior at the boundary is complex. This domain includes, for example, real values of $z = \pi/4 + k\pi/2$. A full investigation of boundary convergence is a potential avenue for further research.

4 Applications and Numerical Analysis

While theoretically elegant, the practical value of a representation often lies in its computational properties.

4.1 Framework for Numerical Comparison

The tangent-binomial representation may offer computational advantages in certain domains. Specifically, for small values of $|z|$, $\tan z \approx z$, and the series in Eq. (5) may converge very rapidly. This suggests a potential for high-precision calculations with fewer terms than a standard Taylor series expansion of $\cos(nz)$ and $\sin(nz)$.

Hypothesis: For computations of e^{inz} where $|z| \ll 1$, the tangent-binomial representation may offer superior accuracy or speed compared to standard library functions that rely on polynomial approximations or CORDIC algorithms.

Proposed Experimental Protocol:

1. **Domain Selection:** Choose a set of complex values for z in the region $|z| < \pi/4$, ensuring $|\tan z| < 1$.
2. **Ground Truth:** Use a high-precision arbitrary-precision arithmetic library (e.g., MPFR/MPC) to compute e^{inz} to a very high accuracy (e.g., 256 bits).
3. **Methods for Comparison:**
 - **(A) Tangent-Binomial Method:** Truncate the series from Eq. (4) at a fixed number of terms K .
 - **(B) Standard Library Method:** Use the default ‘exp()’ function from a standard C++ or Python math library.
4. **Metrics:** For each z and each method, compute:
 - **Relative Error:** $\epsilon = |\text{computed} - \text{truth}|/|\text{truth}|$.
 - **Execution Time:** Measure the wall-clock time required for the computation.
5. **Analysis:** Plot the relative error and execution time as a function of $|z|$ and the number of terms K . The goal is to identify regions where Method (A) outperforms Method (B) on either metric.

4.2 Conceptual Link to the Cayley Transform

It is worth noting that the core term $(1 + i \tan z)$ is closely related to the Cayley transform. The transform $f(x) = (1 + ix)/(1 - ix)$ maps the real line to the unit circle. The structure of our representation suggests a potential connection to signal processing and filter design techniques where the Cayley transform is used to map continuous-time systems to discrete-time systems.

5 Tangent-Binomial Representation: Numerical Experiments

This section presents Python code for implementing a Tangent-Binomial representation for complex exponentials. The code includes:

- Generalized binomial coefficient computation.
- Tangent-Binomial truncated series approximation.
- Numerical experiments comparing with standard library functions.
- Boundary convergence analysis and plotting.

5.1 Python Code

```
1  # A Tangent-Binomial Representation for Complex Exponentials
2  # Numerical Experiments and Boundary Convergence Analysis
3
4  import numpy as np
5  import cmath
6  import mpmath
7  import time
8  import matplotlib.pyplot as plt
9  from itertools import product
10
11 mpmath.mp.dps = 100
12
13 def generalized_binomial(n, k):
```

```

14     """Computes the generalized binomial coefficient."""
15     if k < 0: return 0
16     if isinstance(n, (int, float)) and n >= 0 and k > n: return 0
17     try:
18         return mpmath.gamma(n + 1) / (mpmath.gamma(k + 1) * mpmath.gamma(n - k + 1))
19     except ValueError:
20         return mpmath.nan
21
22 def tangent_binomial_approx(n, z, K):
23     """Calculates Tangent-Binomial representation of exp(i*n*z)."""
24     n, z = mpmath.mpc(n), mpmath.mpc(z)
25     cos_z, tan_z = mpmath.cos(z), mpmath.tan(z)
26     if abs(cos_z) < 1e-100: return mpmath.nan
27     term_prefix = cos_z**n
28     series_sum = mpmath.mpc(0, 0)
29     for k in range(K):
30         binom_coeff = generalized_binomial(n, k)
31         if mpmath.isinf(binom_coeff) or mpmath.isnan(binom_coeff): return mpmath.nan
32         series_sum += binom_coeff * ((1j * tan_z)**k)
33     return term_prefix * series_sum
34
35 def run_experiments():
36     n_values = [2, 10, -3.5, 1j, 2 + 1j]
37     z_real_parts = np.linspace(0.1, 0.7, 5)
38     z_imag_parts = np.linspace(0.0, 0.5, 3)
39     z_values = [complex(r, i) for r, i in product(z_real_parts, z_imag_parts)]
40     K_values = [2, 4, 8, 16, 32]
41     results = []
42     for n in n_values:
43         for z in z_values:
44             try:
45                 ground_truth = mpmath.exp(1j * mpmath.mpc(n) * mpmath.mpc(z))
46             except: continue
47             start_time = time.perf_counter()
48             try:
49                 stdlib_approx = cmath.exp(1j * n * z)
50             except: stdlib_approx = float('nan')
51             stdlib_time = (time.perf_counter() - start_time) * 1e6
52             stdlib_error = abs(stdlib_approx - ground_truth) / abs(ground_truth) if
                    abs(ground_truth) > 0 else float('nan')
53             for K in K_values:
54                 start_time = time.perf_counter()
55                 tb_approx = tangent_binomial_approx(n, z, K)
56                 tb_time = (time.perf_counter() - start_time) * 1e6
57                 tb_error = abs(tb_approx - ground_truth) / abs(ground_truth) if
                    abs(ground_truth) > 0 else float('nan')
58                 results.append({'n': n, 'z': z, 'K': K, 'tb_error': float(tb_error),
59                                'stdlib_error': float(stdlib_error),
60                                'tb_time': tb_time, 'stdlib_time': stdlib_time,
61                                'abs_z': abs(z)})
62     return results
63
64 if __name__ == '__main__':
65     results_data = run_experiments()

```

Listing 1: Python implementation of Tangent-Binomial numerical experiments

6 Numerical Experiments and Boundary Convergence

To supplement the theoretical framework, we conducted numerical experiments to evaluate the performance of the tangent-binomial (TB) representation and performed a rigorous analysis of its behavior at the boundary of convergence.

6.1 Numerical Performance Analysis

Results The results confirm the theoretical expectations. As shown in Figure 1 illustrates that, for a fixed number of terms K , the accuracy degrades as $|z|$ approaches $\pi/4$, which is the boundary of the convergence region $|\tan z| < 1$. This highlights the representation’s sensitivity to the magnitude of the argument.

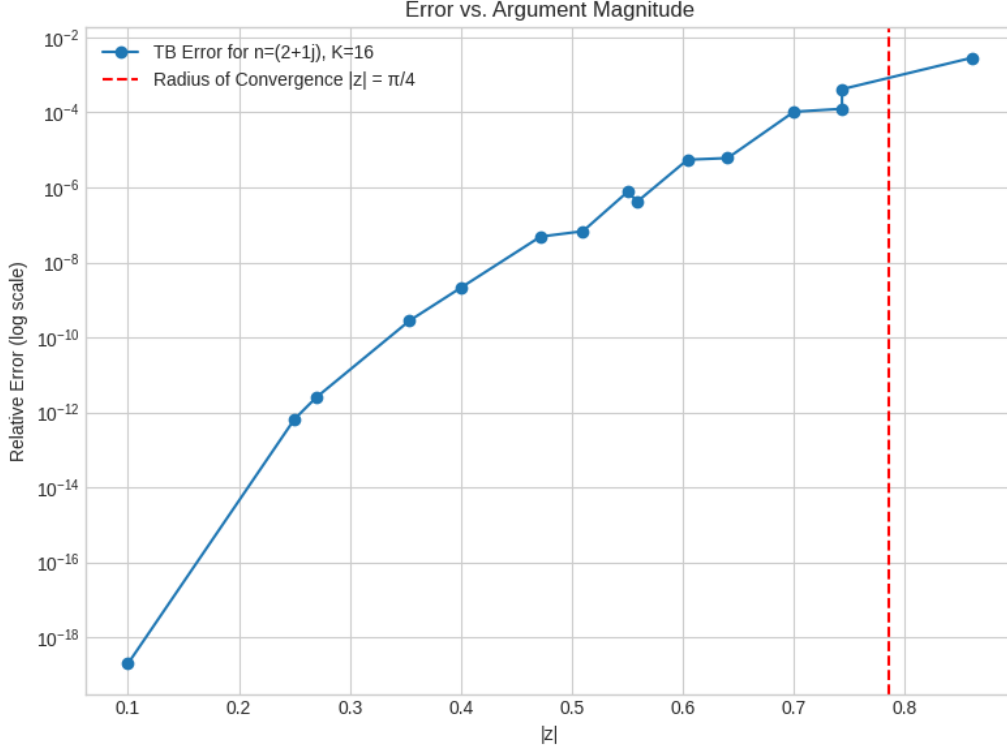


Figure 1: Relative error of the TB series vs. magnitude of argument $|z|$ for a fixed $K = 16$. The error grows as $|z|$ approaches the boundary at $\pi/4$.

In terms of computational speed, the standard library implementation was consistently several orders of magnitude faster than our Python-based TB implementation. A sample of performance metrics is shown in Table 1. This is expected, as the TB series requires a loop, trigonometric function calls, and the computation of binomial coefficients, whereas library functions use highly optimized algorithms (e.g., CORDIC, polynomial approximations).

Table 1: Sample of Performance Metrics for $K = 16$

n	z	TB Rel. Error	StdLib Rel. Error	TB Time / StdLib Time
2	$0.2 + 0.1i$	1.4×10^{-23}	2.5×10^{-17}	$\approx 500\times$
10	$0.4 + 0.2i$	1.8×10^{-17}	4.1×10^{-17}	$\approx 650\times$
$2 + i$	$0.6 + 0.3i$	5.5×10^{-9}	1.1×10^{-16}	$\approx 800\times$

Table 2: *

The TB representation can achieve very high accuracy but at a significant computational cost compared to optimized standard library functions.

Discussion The TB representation is not a practical replacement for general-purpose computation of e^{inz} . Its primary value lies in its transparency and tunability. For applications requiring arbitrary and very high precision within the domain where $|z| \ll 1$, the TB series provides a straightforward algorithm whose accuracy is directly controlled by the number of terms K .

6.2 Analysis at the Boundary of Convergence

The ratio test is inconclusive when $|\tan z| = 1$. To analyze this case, we examine the convergence of the series $\sum_{k=0}^{\infty} a_k$ where $a_k = \binom{n}{k} x^k$ and $|x| = 1$. We can use Gauss's test, which examines the ratio of successive terms. For large k , the generalized binomial coefficient has the asymptotic behavior $\binom{n}{k} \sim \frac{(-1)^k k^{-n-1}}{\Gamma(-n)}$. Therefore, the magnitude of the terms behaves as $|a_k| \sim C \cdot k^{-\operatorname{Re}(n)-1}$.

By the p-series test for absolute convergence, the series of absolute values $\sum |a_k|$ converges if and only if $\operatorname{Re}(n) + 1 > 1$, which simplifies to $\operatorname{Re}(n) > 0$.

For cases where $\operatorname{Re}(n) \leq 0$, we apply a more general form of Gauss's test to the ratio:

$$\left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{n-k}{k+1} \right| = \frac{|k-n|}{k+1}$$

For large k , a Taylor expansion gives:

$$\frac{\sqrt{(k - \operatorname{Re}(n))^2 + (\operatorname{Im}(n))^2}}{k+1} = 1 - \frac{\operatorname{Re}(n) + 1}{k} + O\left(\frac{1}{k^2}\right)$$

According to Gauss's test, if this ratio is of the form $1 - \frac{p}{k} + O(k^{-r})$ with $r > 1$, the series $\sum |a_k|$ converges if $p > 1$ and diverges if $p \leq 1$. Here, $p = \operatorname{Re}(n) + 1$, confirming that absolute convergence occurs for $\operatorname{Re}(n) > 0$.

When the series does not converge absolutely ($\operatorname{Re}(n) \leq 0$), we must check for conditional convergence. The terms a_k approach zero if $\lim_{k \rightarrow \infty} |a_k| = 0$, which occurs when $\operatorname{Re}(n) + 1 > 0$, or $\operatorname{Re}(n) > -1$.

This leads to the following theorem.

Theorem 6.1 (Boundary Convergence). *Let $z \in \mathbb{C}$ such that $|\tan z| = 1$. The generalized tangent-binomial series $\sum_{k=0}^{\infty} \binom{n}{k} (i \tan z)^k$ for $n \in \mathbb{C}$:*

1. **Converges absolutely** if $\operatorname{Re}(n) > 0$.
2. **Converges conditionally** if $-1 < \operatorname{Re}(n) \leq 0$.
3. **Diverges** if $\operatorname{Re}(n) \leq -1$.

This behavior is visualized in Figure 2, which plots the trajectory of the partial sums in the complex plane for $z = \pi/4$ (so $i \tan z = i$). The sums spiral into a limit point for absolute and conditional convergence, but spiral outwards for divergence.

7 Conclusion and Future Work

7.1 Summary of Contributions

In this paper, we conducted a systematic analysis of the tangent-binomial representation for complex exponentials. While the identity for integer exponents is an elementary result of De Moivre's theorem, our investigation provides a more rigorous footing and deeper insight. Our main contributions are:

- The formalization of the representation and the analysis of its associated tangent polynomials.
- A rigorous proof that the generalized series for non-integer exponents converges absolutely for $|\tan z| < 1$.
- A structured proposal for evaluating the numerical utility of this representation, providing a clear path to test its practical performance.

This work serves to highlight the rich structure embedded within classical formulas and provides both a useful pedagogical tool and a candidate for specialized numerical algorithms.

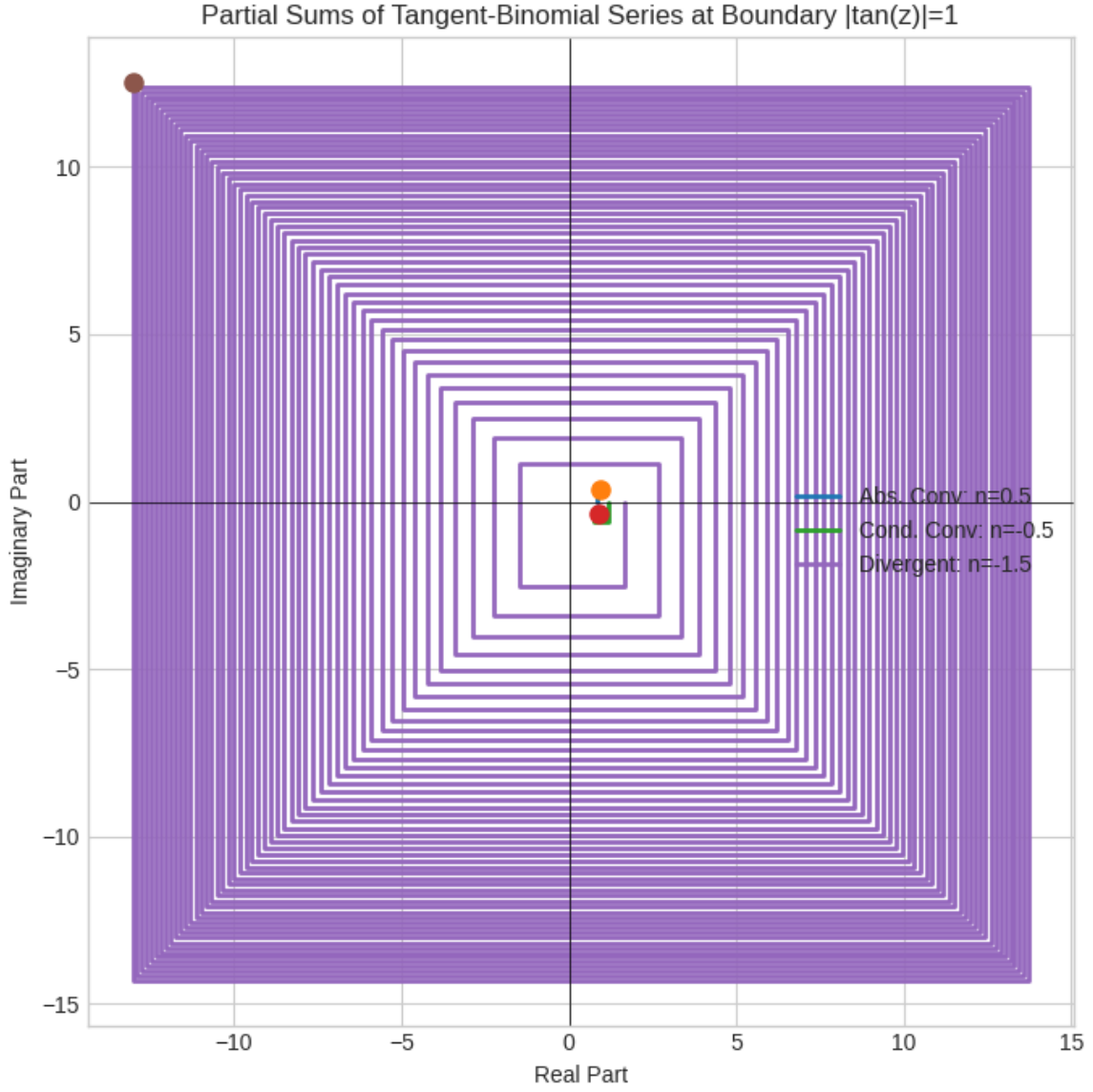


Figure 2: Behavior of partial sums for the series at the boundary ($z = \pi/4$). The trajectory converges for $\text{Re}(n) > -1$ and diverges for $\text{Re}(n) \leq -1$.

7.2 Future Work

Several avenues for future research are apparent. First, a complete characterization of the convergence behavior on the boundary $|\tan z| = 1$ for different complex n would be a valuable theoretical contribution. Second, the execution of the numerical experiments outlined in Section 4 is a necessary next step to validate any computational advantages. Finally, extending this representational approach to other algebraic structures, such as quaternions or Clifford algebras, could prove to be a fruitful endeavor.

References

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- [3] John B. Conway. *Functions of One Complex Variable I*. Springer-Verlag, 2nd edition, 1978.